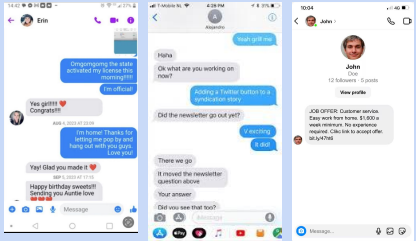
JMessage: Instant Messaging for the TJHSST Community

Sam Stankiewicz & Nidhish Shakya, Dr. Yilmaz Computer Systems Lab 2026

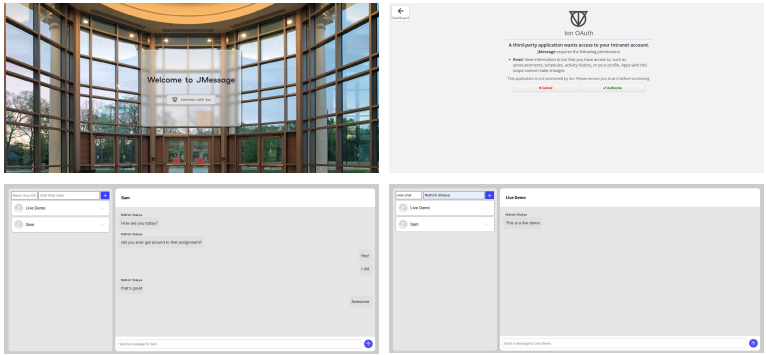
Introduction

While alternate messaging applications are widespread, adaptation to the TJHSST community is limited by several factors that they fail to control for. Widespread phone and computer bans complicate the use of these messaging applications, further exacerbating this issue.



Above are pictured Messenger, iMessage, and Instagram Messages, respectively. All three have inherent shortcomings: the search adaptability of Instagram and Messenger is low, and the barrier to contact in iMessage is high. At any rate, all three solutions are lacking in cross-adaptability to the TJHSST environment. JMessage poses a solution to this immediate deficit.

Results



Our novel solution integrates Ion directory into the traditional messaging interface. This way, the immediate deficit posed by traditional messaging applications is remedied by the incorporation of tools better adapted to the TJHSST community.

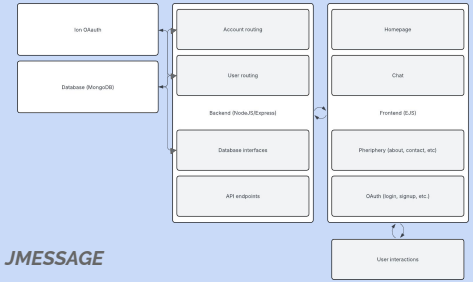
References

https://guides.tjhsst.edu/ion/using-ion-oauth

Safety

One potential benefit of our platform is the increased safety implications of operating a messaging application within a closed, containerized environment as opposed an open collection of accounts from any potential outside actor. By restricting our application exclusively to the TJHSST community, we mitigate this effect and develop a novel gain over traditional messaging.

Methods



JMESSAGE

Our application utilizes a standard NodeJS/ExpressJS application framework with a frontend/backend model. Our methodology is relatively standard to most messaging applications.